

Issues with Bluetooth Sniffer

Privacy Analysis of Augmented Reality Glasses

1 Micro:bit

1.1 What I tried

I flashed a sniffer program (btlejack) onto the microbit (1.4). However, the current issue is that the micro bit is not able to sniff the connection between the phone and glasses. Although the micro bit barely sniffs any connection; at my home , it is not able to sniff any connections, whist at 'dmi' it was able to sniff a couple connections but are not the glasses.

1.2 Powering on

Either - Plug the battery pack into the micro bit. A red light should turn on.
Or - Plug in the micro bit to the laptop using a USB cable. The red light plus another yellow light should turn on.

1.3 Flashing a random program onto the micro bit

I tried this to see if a program was actually being flashed onto the micro:bit since nothing happened when I initially powered on the micro bit.

Plug the micro bit onto the laptop.

One can write code either with MakeCode or on the micro:bit Python Editor.

Download the .hex file (your program) then drag and drop it onto the MICROBIT drive.

An example program that creates a heart can be found in the following link:

<https://microbit.org/get-started/user-guide/reset-your-microbit/>

1.4 Program a BLE sniffer program onto the micro bit

The program that I tried to use and sniff BLE traffic is from

<https://github.com/virtualabs/btlejack>

Btlejack allows sniffing, jamming and hijacking Bluetooth Low Energy devices.

The steps to set up the sniffer:

1. Install the 'btlejack' software with Pip, this would require one to use a virtual environment which can be found in section 3.

```
sudo pip3 install btlejack
```

2. Connect the micro bit device to the laptop with a USB cable.

3. Use the following command which will program every micro bit device connected to the laptop, preparing them for use with Btlejack:

```
btlejack -i
```

4. There should be an output with:

```
Flashing ...  
Flash n device
```

5. Scan existing connections and try to get the access address of active BLE devices with (try to make it so that the glasses and phone are in some kind of interaction):

```
btlejack -s
```

1.5 Re-flash the micro with the original firmware

I did this when I encountered the errors on the micro bit.

Download the .hex file from the link:

<https://microbit.org/get-started/user-guide/firmware/>

The steps are also on the page, but to sum it up:

1. Press and hold on the reset button (same side as the lights, should be on the left side of the yellow light).
2. Plug in the micro bit to the laptop with a USB whilst pressing down on the button.
3. Afterwards, you should see MAINTENANCE, not MICROBIT, on your laptop. Then you can let go of the button.
4. Check one of the three version of the micro bit, this can be found on the bottom right corner (V1, V2.00, V2.20/21).
5. Download the correct version then drag and drop the .hex file onto the drive.
6. Next, wait for the yellow light to stop flashing and when the upgrade is completed, the micro bit will reset.
7. Everything should be done one sees the MICROBIT drive appears on the laptop.

1.6 Errors encountered

1.6.1 Flashing micro bit

The following errors are what I encountered when trying to flash the micro bit:

1. 504 with an unhappy face - indicates that there is an issue during the code upload or execution process. This can be caused if the code is corrupted or if there were issues during file transfer issues.
2. 528 with an unhappy face - indicates a runtime error related to flashing or executing code on the device. This can be caused if the firmware or .hex file flashed was corrupted or incompatible.

1.6.2 Permission issues regarding access

This permission issue is regarding the permission to access the serial port where the micro bit is connected.

The error message looks like

```
...
PermissionError: [Errno 13] Permission denied: '/dev/ttyACMO'
...
```

One can solve this by giving the current user account permission to access the serial port. The steps are as follows:

1. Add the current user to the 'dialout' group:

```
sudo usermod -aG dialout $USER
```

2. Restart or log out and back in for the change to actually take effect.
3. One can verify if the change was made by check if the current username is in the 'dialout' group:

```
groups
```

One should then be able to continue running the bltejack commands.

2 Adafruit

2.1 What I tried

I flashed a sniffer program onto the microbit (2.2). However, the current issue is that the micro bit is not able to sniff the connection between the phone and glasses. Although I see a list of roughly 20 (at my home) different devices, which is more than the micro bits, the glasses are still not included.

2.2 Program a BLE sniffer program onto the Adafruit

The program that I tried to use and sniff BLE traffic is from https://github.com/adafruit/Adafruit_BLESniffer_Python. The python script allows for sniffing Bluetooth Low Energy devices. The steps to set up the sniffer:

1. Clone the git repository onto your laptop

```
git clone https://github.com/adafruit/Adafruit_BLESniffer_Python.git
```

2. Install the python dependencies necessary:

```
pip install -r requirements.txt
```

3. Connect the Adafruit device to the laptop with a USB cable.
4. Determine which serial port to use with the sniffer script:

```
sudo dmesg | grep tty
```

One should see something like:

```
... cp210x converter now attached to ttyUSB0
```

4. Run the sniffer script (replace 'ttyUSB0' with what you see in step 4):

```
python sniffer.py /dev/ttyUSB0
```

2.3 Errors encountered

2.3.1 Not recognizing Adafruit

This is when laptop is not able to find nor identify that Adafruit. On my laptop, there was never a direct indication where I could see that Adafruit was plugged in. So to check if it is connected and the laptop is seeing it, I did the following:

1. Try to see if the device is recognized as a mass storage device:

```
lsblk
```

Mine was never recognized as a mass storage device (like the microbit) but rather connected as a serial device. One can see this by:

```
sudo dmesg | grep tty
```

The device can be recognized with something like the following, which meant it is assigned to (/dev/ttyUSB0):

```
... cp210x converter now attached to ttyUSB0
```

2.3.2 logFilePath wrong

This seems to be an issue with the log file path in the 'Logger.py' script in the python package. The error message looks like

```
File ".../Adafruit_BLESniffer_Python/SnifferAPI/Logger.py", line 22, in <module>
logFilePath = os.path.join(
    ~~~~~
```

```
File "<frozen posixpath>", line 76, in join
TypeError: expected str, bytes or os.PathLike object, not NoneType
```

One can solve this by changing the configuration of the script. The steps are as follows:

1. Open the file 'Logger.py'

```
nano .../Adafruit_BLESniffer_Python/SnifferAPI/Logger.py
```

2. Go to the place where the 'logFilePath' is being defined, which should look something like this:

```
logFilePath = os.path.join(
    os.getenv('APPDATA'), 'Nordic Semiconductor', 'Sniffer'
)
```

3. Replace it with

```
    logFilePath = os.path.join(  
        os.getcwd(), 'Sniffer'  
    )
```

which should change it so that the log file will be saved in the current directory that is being worked on rather than some environment variable.

3 Set Up Virtual environment

The steps to set up a virtual environment:

1. One will need python3, so if it not installed write:

```
sudo apt install python3
```

2. One will also need the venv module, so if it is not installed write:

```
sudo apt install python3-venv
```

3. Create the virtual environment:

```
python3 -m venv myvenv
```

4. Start the virtual environment:

```
source myvenv/bin/activate
```

5. Then one should see '(myvenv)' in front of the terminal prompt.

6. To stop the virtual environment:

```
deactivate
```

4 Finding the MAC address of devices

4.1 Ray Ban Meta glasses

An application on the Android phone called 'nRF Connect' was used.

Once in the app, turn on the glasses.

After the light turns green on the notification LED on the glasses, refresh the 'scanner' page to find devices.

There should be a list of devices, check the 'Company' to see if the devices found are the glasses.

The 'Company' for the glasses should be 'Meta Platforms'.

The MAC address found for the glasses is 98:59:49:34:7A:8C.

4.2 RedMi Note 8 phone

In the settings app, go to 'About phone', 'All specs', 'Status'.

In there the MAX address found is b8:3b:cc:da:73:a8.